

# Four Generative Models for Image Restoration and Face-to-Sketch Synthesis: Autoencoders, Hard and Soft Routing, and a Style-Conditioned cGAN

Muhammad Harris  
Student ID i232535  
Generative AI, Assignment 1  
i232535@isb.nu.edu.pk

**Abstract**—We study four related generative systems. A universal denoising autoencoder restores images corrupted by salt-and-pepper noise, Gaussian blur, or rectangular occlusion without being told which corruption is present. A corruption classifier routes each image to one of three specialist autoencoders (hard routing), and a soft mixture-of-experts variant blends the same specialists with learned, differentiable weights. A conditional GAN with a U-Net generator and a PatchGAN discriminator converts face photographs into sketches in one of three styles. Every model is tuned with Optuna. The system serves all four models through one browser application with a FastAPI backend and a React frontend, with models exported to ONNX. This report documents the research behind each architectural choice, the alternatives we rejected and why, the hyperparameter studies, and the evaluation protocol.

Headline results on the official test sets are as follows. The universal autoencoder improves on noisy and occluded inputs, by up to 9.9dB on salt-and-pepper and 7.6dB on occlusion. It does not beat a  $3\times 3$  median filter on salt-and-pepper noise, and it makes blurred images worse. The classifier identifies the corruption with 98.7% accuracy (macro F1 0.979). The soft mixture routes salt-and-pepper to its expert with 0.95 or more of the weight at medium and high severity, but routes mild blur mainly to the identity branch. The face-to-sketch generator reaches a test L1 of 0.097 and SSIM of 0.564, with Style 2 the hardest.

**Index Terms**—denoising autoencoder, mixture of experts, hard routing, conditional GAN, PatchGAN, U-Net, Optuna, ONNX

## I. INTRODUCTION

Photographs are often damaged by noise, blur, or occlusion, and a common approach is to learn a single restoration network. A different approach is to first diagnose the damage and then dispatch the image to a specialist. This report compares the two. We also study whether soft, differentiable routing can combine specialists better than a hard switch. The fourth task is a different problem: translating a face photograph into a stylized sketch with a conditional generative adversarial network (cGAN).

The assignment was designed to test research and engineering together. Each design decision in this report is justified by the literature and by experiments, and each alternative we considered is listed with the reason it was not chosen.

## II. RELATED WORK

**Denoising autoencoders.** Vincent et al. showed that training an autoencoder to reconstruct a clean input from a corrupted version yields representations that are robust to the corruption [1]. This is the training objective used in Task 1.

**Upsampling artifacts.** Transposed convolutions can produce checkerboard patterns because of uneven kernel overlap. Odena et al. show that separating resizing from convolution avoids the problem [2]. We use this upsample-then-convolve pattern in every decoder.

**Losses for restoration.** Zhao et al. compare pixel losses and perceptual losses and find that combining L1 with a structural term gives better results than L2 alone [3]. The structural similarity index (SSIM) was introduced by Wang et al. [4].

**Normalization.** Batch normalization depends on batch statistics, which become noisy at small batch sizes. Group normalization removes that dependence [5]. Instance normalization is common in image-to-image networks [6].

**Mixture of experts.** Jacobs et al. introduced adaptive mixtures of local experts, in which a gating network assigns weights to experts [7]. Shazeer et al. scaled this to large networks [8]. A known failure mode is routing collapse, where one expert receives almost all inputs; Task 3 uses a balance regularizer against it.

**Conditional image translation.** Isola et al. introduced pix2pix, a conditional GAN with a U-Net generator and a PatchGAN discriminator, for paired image translation [9]. Our Task 4 model follows this design. Perez et al. propose feature-wise linear modulation (FiLM) as a general conditioning mechanism [10], which we considered and rejected (Section VI).

**Hyperparameter optimization.** Optuna uses a define-by-run API and pruning of unpromising trials [11].

**Datasets.** The Oxford-IIIT Pet dataset [12] provides the images for Tasks 1–3. The FS2K dataset [13] provides the photo-sketch pairs for Task 4.

TABLE I  
CORRUPTION CONFIGURATION. TRAINING SAMPLES SEVERITY AT  
RANDOM; TEST USES THE FIXED LEVELS.

| Condition       | Training                                     | Test low   | Test medium / high          |
|-----------------|----------------------------------------------|------------|-----------------------------|
| Clean           | none                                         | none       | none                        |
| Salt-and-pepper | $p \sim U(0.02, 0.15)$                       | $p = 0.03$ | $0.08 / 0.13$               |
| Gaussian blur   | $k \in \{3, 5, 7\}, \sigma \sim U(0.5, 2.5)$ | $(3, 0.7)$ | $(5, 1.5) / (7, 2.5)$       |
| Occlusion       | 1–3 boxes, 10–35% area                       | 1 box, 10% | 2 boxes, 20% / 3 boxes, 35% |

### III. DATASET PREPARATION

#### A. Oxford-IIIT Pet (Tasks 1–3)

The official training and validation list (`trainval.txt`, 3,680 images) is the development set. We split it 80/20 with seed 42, giving 2,944 training and 736 validation images. The official test list (`test.txt`, 3,669 images) is used only for evaluation. All images are converted to RGB and resized to  $128 \times 128$ . Corruptions are generated at runtime during training and stored in fixed manifests for validation and testing. Table I gives the definitions.

#### B. FS2K (Task 4)

FS2K contains 2,104 photo-sketch pairs in three styles [13]. We use the official training file (1,058 pairs) and the official test file (1,046 pairs). From the training file we hold out 15% for validation (159 pairs), stratified by style with seed 42. The split is 899 training, 159 validation, and 1,046 test images. Photos and sketches are resized to  $128 \times 128$ , and augmentation (horizontal flip, random crop) is applied to each pair with the same random draw. Style labels come from the `style` field of the annotations, because the folder structure does not match the styles.

### IV. TASK 1: UNIVERSAL DENOISING AUTOENCODER

#### A. Architecture and alternatives

Table II lists each design choice, the options considered, and the reason for the choice. The resulting model is fully convolutional. There are no skip connections, so the bottleneck must carry the information needed for reconstruction.

The bottleneck was tested at two depths and two widths: depth 4 gives an  $8 \times 8$  map and depth 5 gives  $4 \times 4$ . Channels double at each stage from a base width. The compression ratio is the input size divided by the bottleneck size,  $49,152 / (H \cdot W \cdot C)$ . For example,  $8 \times 8 \times 256$  gives  $3 \times$  compression and  $8 \times 8 \times 512$  gives  $1.5 \times$ .

#### B. Optuna study

The study ran 30 trials, each for three epochs on the training split, with median pruning. The search space was: learning rate in  $\{10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}\}$ ; batch size in  $\{8, 16, 32, 64\}$ ; spatial size  $\{4, 8\}$ ; base channels  $\{16, 32, 64\}$ ; dropout in  $[0, 0.3]$ ;  $\alpha \in [0.5, 0.95]$ ; activation in  $\{\text{ReLU}, \text{LeakyReLU}\}$ . Table III lists the trials that matter for the decision.

Trial 26 had the lowest validation loss, but its bottleneck compresses the input by only  $1.5 \times$ . Trial 17 is 0.0923, about

11% higher, and compresses by  $3 \times$ . It keeps  $3 \times$  compression. We chose trial 17. The loss increase is small, and the bottleneck is a meaningful compression of the input, which the assignment requires. Trial 13 has stronger compression ( $12 \times$ ) but a loss 59% higher than trial 17. We judged that too large a cost.

The selected hyperparameters are: learning rate  $3 \times 10^{-4}$ , batch size 16, spatial size 8, base channels 32, dropout 0.158,  $\alpha = 0.927$ , LeakyReLU.

#### C. Training and evaluation

The final model is trained for 50 epochs on the training split with the selected settings. Evaluation uses the official test list and the fixed test grid: each image is tested with the clean input and with three severities of each corruption, ten entries per image. Results are reported as L1, PSNR, and SSIM per corruption and severity, and include twelve examples (clean, corrupted, restored, absolute error map).

Each entry is scored three ways: the model’s restored output, the corrupted input itself (no restoration), and, for salt-and-pepper only, a  $3 \times 3$  median filter. Gains are the model minus the baseline, so a positive gain means the model improves on that baseline.

The results show three distinct patterns. For salt-and-pepper noise, the model improves on the noisy input by up to 9.9 dB, but a simple median filter is better by 5 to 8 dB at every severity. For blur, the model’s output scores below the blurred input at every severity, so restoration makes the image worse. For occlusion, the model improves on the input by 5 to 8 dB, but its output reaches only 19 to 22 dB. Even on clean images, the model reaches only 23.8 dB and an SSIM of 0.70, which means its output is smooth even when nothing needs fixing.

[TODO: Task 1 training and validation loss curves]

TABLE II  
TASK 1 DESIGN DECISIONS.

| Decision         | Options considered                          | Chosen                                       | Reason                                                                                 |
|------------------|---------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------|
| Downsampling     | max pooling; strided convolution            | strided conv, $k=3$ , $s=2$                  | the network learns the downsampling instead of discarding information with a fixed max |
| Upsampling       | transposed conv; upsample + conv            | upsample + conv                              | avoids checkerboard artifacts [2]                                                      |
| Normalization    | BatchNorm; GroupNorm; none                  | GroupNorm (8 groups)                         | batch size is searched over 8–64; GroupNorm does not depend on batch statistics [5]    |
| Activation       | ReLU; LeakyReLU(0.2)                        | searched by Optuna                           | both are cheap; the search decides                                                     |
| Bottleneck       | flattened dense vector; spatial feature map | spatial feature map                          | preserves spatial layout; no large dense layer                                         |
| Skip connections | none; limited skips                         | none                                         | the assignment requires a genuine bottleneck                                           |
| Output           | linear; sigmoid                             | sigmoid                                      | images are in $[0, 1]$                                                                 |
| Loss             | L1; L2; L1+SSIM                             | $\alpha L_1 + (1 - \alpha)(1 - \text{SSIM})$ | L1 gives accuracy, SSIM gives structure [3], [4]                                       |

TABLE III  
TASK 1 OPTUNA TRIALS (VALIDATION LOSS, LOWER IS BETTER). TRIAL 26 WAS THE UNCONSTRAINED WINNER. TRIAL 17 WAS THE BEST CONFIGURATION WITH  $3\times$  OR GREATER COMPRESSION.

| Trial | Bottleneck               | Compression | Activation | Val. loss |
|-------|--------------------------|-------------|------------|-----------|
| 26    | $8 \times 8 \times 512$  | $1.5\times$ | ReLU       | 0.0830    |
| 15    | $8 \times 8 \times 512$  | $1.5\times$ | LeakyReLU  | 0.0933    |
| 17    | $8 \times 8 \times 256$  | $3\times$   | LeakyReLU  | 0.0923    |
| 3     | $8 \times 8 \times 128$  | $6\times$   | LeakyReLU  | 0.1501    |
| 13    | $4 \times 4 \times 256$  | $12\times$  | LeakyReLU  | 0.1472    |
| 7     | $4 \times 4 \times 1024$ | $3\times$   | ReLU       | 0.1794    |

#### D. Failure cases

We identify four failure cases from the test results. Each is backed by the numbers above. The explanations are hypotheses, and the example images below should be inspected to confirm them.

**Failure 1: the model ignores its blurred input.** At low blur severity, the model scores 23.9 dB, while the blurred input scores 34.2 dB. The model’s output looks much the same regardless of input, which suggests it has learned a generic smooth image. Our hypothesis is that the bottleneck and the training objective favour smooth outputs, so the model discards the fine detail that a mild blur leaves intact.

**Failure 2: noise removal is weaker than a median filter.** At high salt-and-pepper severity, the model reaches 23.0 dB, a gain of 9.9 dB over the noisy input, but the median filter reaches 29.0 dB. A local median filter handles impulse noise directly, which the learned model does not match. We hypothesise that the L1 and SSIM loss mix does not penalise the model enough for keeping isolated noise pixels.

**Failure 3: the reconstruction ceiling.** On clean images, the model reaches only 23.8 dB and an SSIM of 0.70. Every corrupted entry is restored to at most this quality, so the ceiling limits all the other results. We hypothesise that the  $8 \times 8 \times 256$  bottleneck ( $3\times$  compression) discards fur texture and edges that a larger bottleneck would keep. The Optuna search favoured the  $1.5\times$  bottleneck on loss alone (Table III), which points the same way.

**Failure 4: large occlusions cannot be filled.** At high occlusion severity, the input covers up to 35% of the image, and the model reaches 19.1 dB and SSIM 0.60. The model can’t reconstruct content hidden behind a large black box from the compressed code. This is the one corruption where restoration improves on the input, but the result is still far below the clean reconstruction.

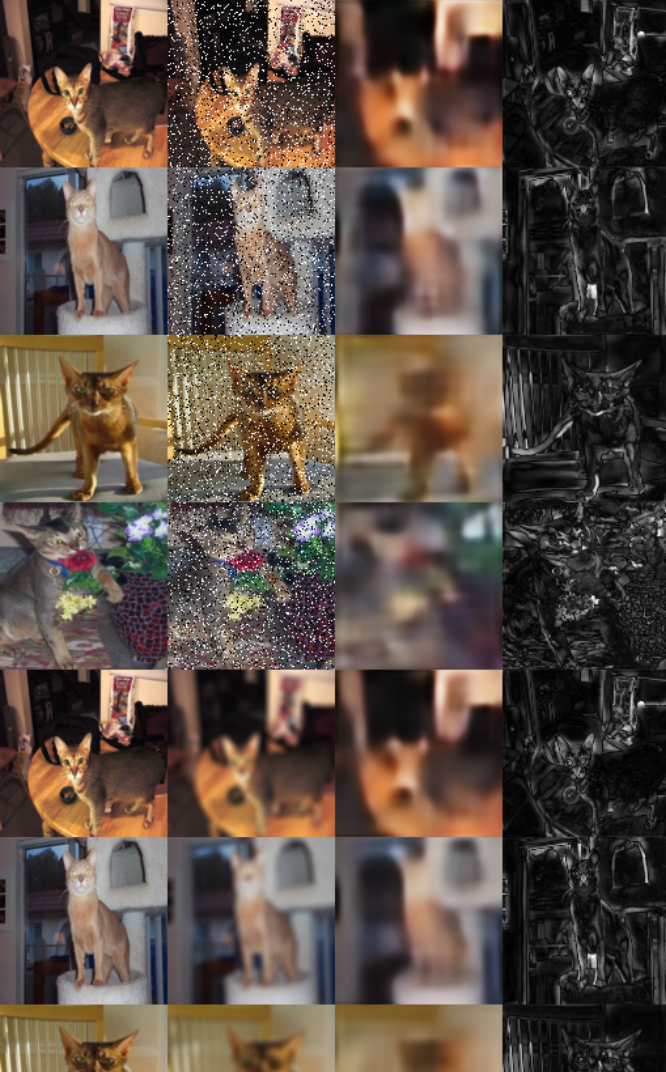


TABLE IV

TASK 1 TEST RESULTS ON THE OFFICIAL TEST LIST, 3,669 IMAGES PER ROW. GAIN = MODEL MINUS INPUT (OR MINUS MEDIAN FILTER, WHERE GIVEN). CLEAN ROWS HAVE NO CORRUPTION, SO THEIR INPUT EQUALS THE TARGET AND GAINS ARE NOT DEFINED.

| Corruption      | Severity | PSNR (dB) |       |        |        | SSIM  |       |        |        |
|-----------------|----------|-----------|-------|--------|--------|-------|-------|--------|--------|
|                 |          | Model     | Input | Gain   | Median | Model | Input | Gain   | Median |
| Clean           | none     | 23.82     | —     | —      | —      | 0.696 | 1.000 | —      | —      |
| Salt-and-pepper | low      | 23.72     | 20.15 | +3.57  | 32.19  | 0.689 | 0.613 | +0.076 | 0.926  |
| Salt-and-pepper | medium   | 23.47     | 15.89 | +7.59  | 31.12  | 0.676 | 0.356 | +0.319 | 0.918  |
| Salt-and-pepper | high     | 23.03     | 13.16 | +9.87  | 28.99  | 0.657 | 0.224 | +0.434 | 0.899  |
| Blur            | low      | 23.89     | 34.20 | −10.31 | —      | 0.695 | 0.962 | −0.266 | —      |
| Blur            | medium   | 23.90     | 28.14 | −4.24  | —      | 0.690 | 0.855 | −0.165 | —      |
| Blur            | high     | 23.68     | 25.37 | −1.69  | —      | 0.676 | 0.755 | −0.080 | —      |
| Occlusion       | low      | 22.10     | 16.78 | +5.32  | —      | 0.665 | 0.880 | −0.215 | —      |
| Occlusion       | medium   | 20.71     | 13.67 | +7.04  | —      | 0.636 | 0.777 | −0.142 | —      |
| Occlusion       | high     | 19.06     | 11.48 | +7.58  | —      | 0.596 | 0.653 | −0.057 | —      |

## V. TASK 2: CORRUPTION CLASSIFICATION AND HARD-ROUTED SPECIALISTS

### A. Classifier

The classifier uses the same down-block as Task 1 (strided convolution, GroupNorm, activation), four stages, then global average pooling and a linear layer with four outputs. We chose global average pooling over flattening because the decision is whether a corruption is present anywhere, not where it is. Batches are balanced across the four classes with a custom sampler, so the classifier cannot learn a bias toward one class.

The Optuna search tuned learning rate, batch size, base channels, dropout, and weight decay. The selected values are: learning rate  $3 \times 10^{-4}$ , batch size 16, base channels 32, dropout 0.018, weight decay  $2.0 \times 10^{-5}$ .

### B. Specialists

All three specialists use the same autoencoder as Task 1. The architecture was chosen with one shared search on pooled corrupted data, then three copies were trained independently, one per corruption.

The shared search ran 20 trials of three epochs each. The unconstrained best trial (#10) had a  $8 \times 8 \times 512$  bottleneck, which compresses only  $1.5\times$ . Within the four bottleneck options we set in the design, the best trial was #12, with an  $8 \times 8 \times 256$  bottleneck and validation loss 0.1196. We selected trial 12. It is 32% higher in loss than trial 10, and it keeps the  $3\times$  compression required by the design. Table V lists the selected configuration.

### C. Routing and evaluation

At inference the classifier chooses an expert. Clean predictions bypass the experts. We evaluate two modes. In *oracle routing* the true label chooses the expert, which isolates specialist quality. In *predicted routing* the classifier chooses, which measures the full system. The difference between the two is the cost of classification errors. We also collect the images where a misclassification most increased the error relative to oracle routing.

TABLE V

TASK 2 SPECIALIST SEARCH, SELECTED CONFIGURATION (TRIAL 12).

| Parameter                | Value                   |
|--------------------------|-------------------------|
| Bottleneck               | $8 \times 8 \times 256$ |
| Base channels            | 32                      |
| Learning rate            | $3 \times 10^{-4}$      |
| Batch size               | 32                      |
| Dropout                  | 0.280                   |
| $\alpha$                 | 0.870                   |
| Activation               | LeakyReLU               |
| Validation loss (pooled) | 0.1196                  |

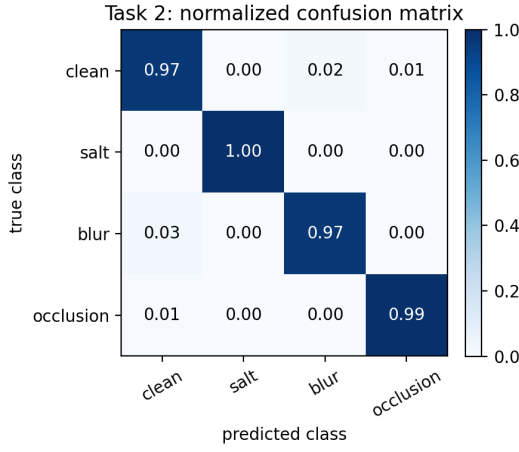
TABLE VI

TASK 2 CLASSIFIER RESULTS ON THE OFFICIAL TEST SET.

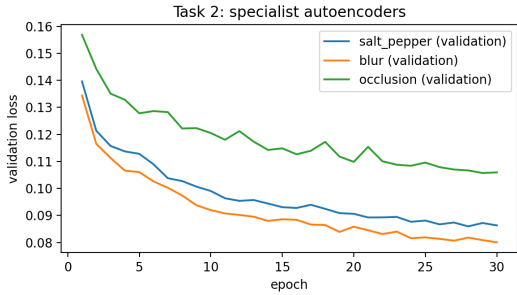
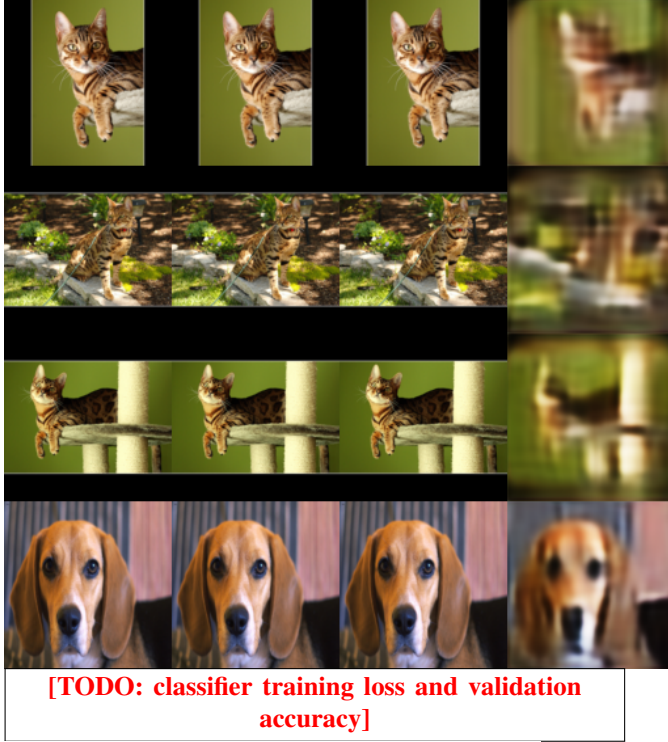
| Class           | Precision | Recall | F1    |
|-----------------|-----------|--------|-------|
| Clean           | 0.905     | 0.972  | 0.937 |
| Salt-and-pepper | 0.999     | 1.000  | 1.000 |
| Blur            | 0.993     | 0.971  | 0.982 |
| Occlusion       | 0.998     | 0.994  | 0.996 |
| Macro average   | 0.974     | 0.984  | 0.979 |

The classifier was evaluated on the official test set (3,669 images, 36,690 test entries, with every corruption at all three severities). Table VI gives the per-class and macro scores, and the figure below shows the normalized confusion matrix.

Overall accuracy is 98.7%. Clean is the weakest class: its precision is 0.905 because some blurred images are read as clean. The confusion matrix shows the same pattern. Two percent of clean images are assigned to blur, and 2.8% of blurred images are assigned to clean. Mild blur is the corruption closest to a clean image, so these are the errors to expect. Salt-and-pepper and occlusion are almost never confused with anything



else.



## VI. TASK 3: SOFT MIXTURE-OF-EXPERTS RESTORATION

### A. Architecture and alternatives

Hard routing sends each image to one expert and cannot handle mixed corruptions or uncertain classifications. We replace it with a soft mixture: a gating network produces weights  $w = \text{softmax}(G(\tilde{x})/\tau)$  over the identity branch and the three

TABLE VII  
TASK 3 DESIGN DECISIONS.

| Decision                | Options considered                   | Chosen                                                                                |
|-------------------------|--------------------------------------|---------------------------------------------------------------------------------------|
| Gate                    | new network; Task 2 classifier       | Task 2 classifier                                                                     |
| Initialization          | random; pretrained                   | pretrained from Task 2                                                                |
| Training                | joint from start; warm-up then joint | warm-up (experts frozen), then joint                                                  |
| Routing                 | hard; softmax with temperature       | softmax with temperature $\tau$                                                       |
| Collapse guard          | none; balance loss                   | balance loss                                                                          |
| Loss weight $\lambda_s$ | tuned freely; $1 - \lambda_1$        | $\sum_k (\bar{w}_k - \frac{1}{4})^2$<br>$1 - \lambda_1$ (keeps weights complementary) |

TABLE VIII  
TASK 3 OPTUNA RESULTS: FIVE BEST TRIALS BY VALIDATION LOSS (LOWER IS BETTER).

| Trial | Val loss | lr                    | Temperature $\tau$ | $\lambda_c$ | $\lambda_b$ | $\lambda_1$ |
|-------|----------|-----------------------|--------------------|-------------|-------------|-------------|
| 16    | 0.0583   | $1.15 \times 10^{-5}$ | 1.29               | 0.014       | 0.020       | 0.882       |
| 14    | 0.0600   | $9.11 \times 10^{-6}$ | 1.79               | 0.018       | 0.047       | 0.880       |
| 11    | 0.0618   | $9.80 \times 10^{-5}$ | 1.00               | 0.018       | 0.051       | 0.890       |
| 12    | 0.0619   | $4.34 \times 10^{-5}$ | 1.65               | 0.018       | 0.091       | 0.890       |
| 18    | 0.0628   | $3.21 \times 10^{-6}$ | 1.46               | 0.011       | 0.075       | 0.870       |

experts, and the output is  $\hat{x} = \sum_k w_k A_k(\tilde{x})$ . Table VII lists the alternatives.

### B. Training and Optuna

Training has two stages. In warm-up the experts are frozen and only the gate learns, so it learns routing against good experts. In joint fine-tuning everything is unfrozen at a lower learning rate. The loss is  $\mathcal{L} = \lambda_1 L_1 + \lambda_s (1 - \text{SSIM}) + \lambda_c L_{CE} + \lambda_b L_{\text{balance}}$ , starting from 0.8/0.2/0.1/0.01. The Optuna search covers joint learning rate in  $[10^{-6}, 10^{-4}]$ , temperature in  $[0.3, 3]$ ,  $\lambda_c \in [0.01, 0.5]$ ,  $\lambda_b \in [0.001, 0.1]$ , and  $\lambda_1 \in [0.5, 0.9]$ . Trials are pruned when two or more experts become inactive. The search ran 20 trials. Each trial used one warm-up epoch and three joint-fine-tuning epochs on the training split. Nineteen trials completed and one was pruned. Trial 16 gave the lowest validation loss (0.0583), and Table VIII lists the five best trials.

The selected configuration is trial 16. Two of its values sit near the edges of their ranges:  $\lambda_1$  (0.882 against a maximum of 0.9) and  $\lambda_c$  (0.014 against a minimum of 0.01). The top trials are close, so the ranking among them is not reliable, as in the other searches. The final model was trained for 20 joint epochs. Its validation loss fell from 0.0585 to 0.0572 and then levelled off (the figure below).

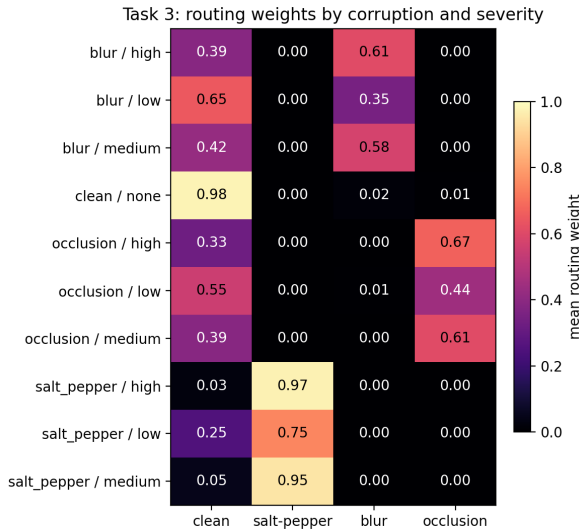
### C. Routing analysis

We report the mean routing weights for each true corruption type and severity, and check for dead experts (mean weight below 0.05) and for experts that dominate unrelated inputs. Table IX gives the mean routing weights for each corruption and severity, on the 300-image test subset. The heatmap and the examples below show the same weights.

TABLE IX  
TASK 3 MEAN ROUTING WEIGHTS BY TRUE CORRUPTION AND SEVERITY  
(300 OFFICIAL TEST IMAGES PER GROUP).

| Corruption      | Severity | Clean | Salt  | Blur  | Occlusion |
|-----------------|----------|-------|-------|-------|-----------|
| Clean           | none     | 0.978 | 0.000 | 0.016 | 0.006     |
| Salt-and-pepper | low      | 0.253 | 0.746 | 0.000 | 0.001     |
| Salt-and-pepper | medium   | 0.051 | 0.949 | 0.000 | 0.000     |
| Salt-and-pepper | high     | 0.027 | 0.973 | 0.000 | 0.000     |
| Blur            | low      | 0.648 | 0.000 | 0.350 | 0.002     |
| Blur            | medium   | 0.423 | 0.000 | 0.576 | 0.001     |
| Blur            | high     | 0.390 | 0.000 | 0.608 | 0.001     |
| Occlusion       | low      | 0.552 | 0.001 | 0.005 | 0.442     |
| Occlusion       | medium   | 0.389 | 0.001 | 0.003 | 0.608     |
| Occlusion       | high     | 0.327 | 0.000 | 0.002 | 0.671     |

No expert is dead: the lowest mean weight across all inputs is 0.156, for the blur expert, above the 0.05 threshold. No expert dominates unrelated inputs: averaged over each corruption group, no expert other than the identity branch receives more than 0.6 of the weight on a corruption it does not match. Salt-and-pepper is routed almost perfectly. The salt expert takes 0.95 or more of the weight at medium and high severity. Blur is different. At low severity the identity branch takes 0.65 of the weight, so mild blur is treated as nearly clean. This matches the confusion between blur and clean in the classifier (Table VI). Occlusion is split between the occlusion expert and the identity branch, and the occlusion share rises from 0.44 at low severity to 0.67 at



high.

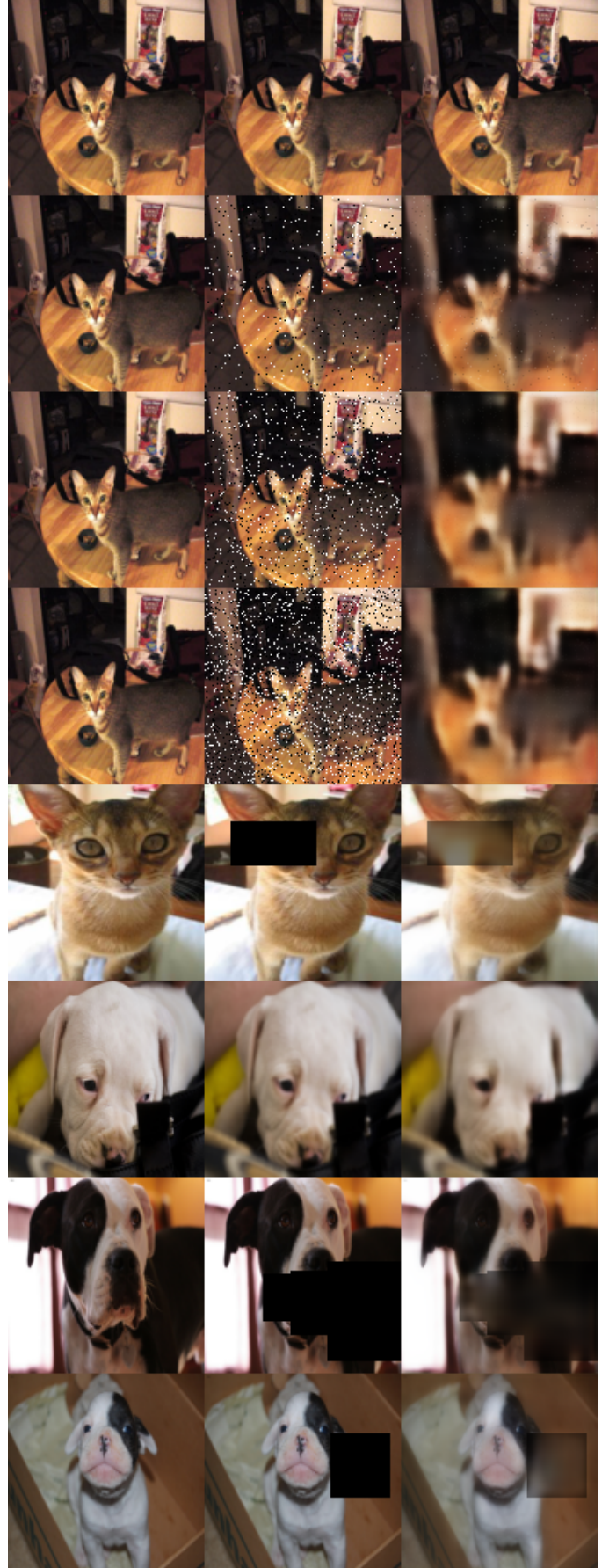


TABLE X  
TASK 3 RECONSTRUCTION QUALITY (300 OFFICIAL TEST IMAGES PER GROUP).

| Corruption      | Severity | L1     | PSNR (dB) | SSIM  |
|-----------------|----------|--------|-----------|-------|
| Clean           | none     | 0.0009 | 62.10     | 0.999 |
| Salt-and-pepper | low      | 0.0380 | 24.54     | 0.725 |
| Salt-and-pepper | medium   | 0.0474 | 23.29     | 0.675 |
| Salt-and-pepper | high     | 0.0504 | 22.83     | 0.655 |
| Blur            | low      | 0.0231 | 29.18     | 0.907 |
| Blur            | medium   | 0.0362 | 25.45     | 0.777 |
| Blur            | high     | 0.0409 | 24.41     | 0.720 |
| Occlusion       | low      | 0.0375 | 22.84     | 0.856 |
| Occlusion       | medium   | 0.0507 | 21.48     | 0.758 |
| Occlusion       | high     | 0.0651 | 19.65     | 0.683 |



## D. Results

Table X gives the reconstruction quality of the soft mixture for each group. These values use the first 300 official test images, so they are not directly comparable to the full-set Task 1 values. A direct comparison needs the full test set.

The clean row is near-identity: the identity branch takes 0.98 of the weight, so the output almost equals the input. For blur, the mixture does not improve on the input. At low severity it reaches 29.2 dB, while the unrestored blurred input scored 34.2 dB in the Task 1 evaluation. The two numbers come from different subsets, so this should be confirmed on the full set, but the blur result points the same way as Task 1: doing nothing beats the restoration. For salt-and-pepper, the mixture’s PSNR (22.8 dB at high severity) is again below the median-filter baseline reported for Task 1, which reached 29.0 dB. The gating does not change that conclusion.

## VII. TASK 4: STYLE-CONDITIONED FACE-TO-SKETCH CGAN

### A. Style conditioning: alternatives and choice

Three ways of conditioning the generator on style were considered. (A) *Input concatenation*: a learned embedding for each style is repeated over the image and concatenated to the input, in both the generator and the discriminator. (B) *Bottleneck injection*: the embedding is applied only at the U-Net bottleneck. (C) *FiLM*: feature-wise scaling and shifting at each stage [10].

We chose (A) and did not use (B). In a U-Net, the skip connections carry encoder features directly to the decoder without passing through the latent space. A style signal applied only at the bottleneck could therefore be skipped, with the

TABLE XI  
TASK 4 DESIGN DECISIONS.

| Decision         | Options considered      | Chosen                         |
|------------------|-------------------------|--------------------------------|
| Generator        | encoder-decoder; U-Net  | U-Net (skips needed for edges) |
| Depth            | 4, 5, 6 stages          | 5 stages (4×4 bottleneck)      |
| Normalization    | BatchNorm; InstanceNorm | InstanceNorm                   |
| Discriminator    | whole-image; PatchGAN   | PatchGAN, RF 34 px             |
| Adversarial loss | BCE; least squares      | BCE with logits (assignment)   |
| Output           | sigmoid; tanh           | tanh on $[-1, 1]$              |

TABLE XII  
TASK 4 OPTUNA RESULTS: FIVE BEST TRIALS BY VALIDATION L1 (LOWER IS BETTER).

| Trial | Val L1 | $\text{lr}_G$        | $\text{lr}_D$        | Batch | Base | Dropout | Style |
|-------|--------|----------------------|----------------------|-------|------|---------|-------|
| 25    | 0.0928 | $2.3 \times 10^{-4}$ | $3.5 \times 10^{-4}$ | 8     | 64   | 0.29    |       |
| 23    | 0.0934 | $1.5 \times 10^{-4}$ | $3.1 \times 10^{-4}$ | 8     | 64   | 0.48    |       |
| 21    | 0.0935 | $1.8 \times 10^{-4}$ | $2.9 \times 10^{-4}$ | 8     | 64   | 0.38    |       |
| 16    | 0.0947 | $3.4 \times 10^{-4}$ | $2.4 \times 10^{-4}$ | 8     | 64   | 0.37    |       |
| 24    | 0.0952 | $1.1 \times 10^{-4}$ | $3.4 \times 10^{-4}$ | 8     | 64   | 0.44    |       |

decoder receiving the style only through features that bypass the bottleneck. Input concatenation gives every layer the style from the start. C is the most expressive, but with three styles and 899 training images the added machinery is hard to justify.

### B. Architecture and alternatives

The generator is a U-Net with five downsampling stages, from 128 to a  $4 \times 4$  bottleneck, so the skip connections keep detail at 64, 32, 16, and 8 pixels. Decoders use upsample-then-convolve and InstanceNorm, with dropout in the first three blocks. The output is tanh on  $[-1, 1]$ . We chose InstanceNorm over BatchNorm because batch sizes from 8 to 32 are searched. The discriminator is a PatchGAN with two stride-2 layers and two stride-1 layers. Its receptive field is 34 pixels, which matches the relative patch size of pix2pix at this resolution [9]. Table XI summarizes the remaining options.

### C. Loss and training

The generator minimizes  $\mathcal{L}_G = \mathcal{L}_{adv} + \lambda_{L1}\mathcal{L}_{L1}$  and the discriminator minimizes binary cross-entropy on real and generated pairs. We log the discriminator real loss, discriminator fake loss, generator adversarial loss, generator L1 loss, and validation L1 and SSIM. Optuna searches generator and discriminator learning rates ( $[10^{-4}, 4 \times 10^{-4}]$ , log scale), batch size {8, 16, 32}, base channels {32, 64}, dropout [0, 0.5], style embedding size {8, 16, 32}, and  $\lambda_{L1} \in [10, 200]$ . Trials run 15 epochs; the winner is retrained for 200 epochs. The search ran 20 trials of 15 epochs each, using the full official training split (899 training and 159 validation images). The test set was not used. Median pruning stopped 6 of the 20 trials. Trials 0–5 of the study failed on a filename error before any training, so they are excluded. Table XII lists the five best trials.

TABLE XIII  
TASK 4 TEST RESULTS ON THE OFFICIAL FS2K TEST SPLIT, PER SKETCH  
STYLE.

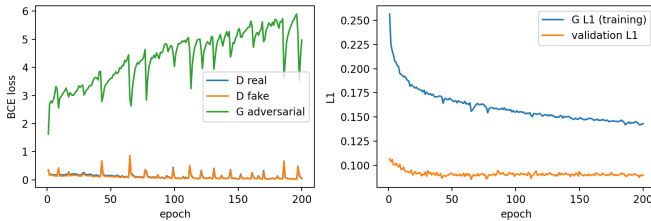
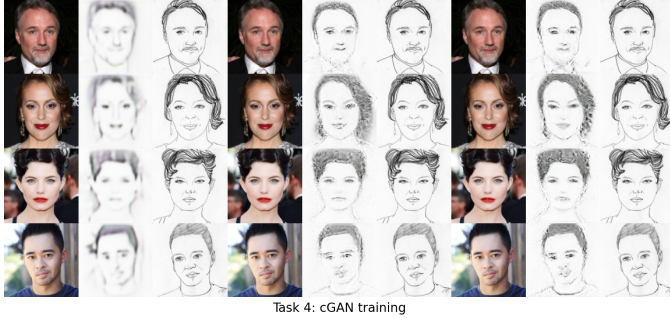
| Style      | Test pairs | L1     | SSIM  |
|------------|------------|--------|-------|
| Style 1    | 619        | 0.0726 | 0.611 |
| Style 2    | 381        | 0.1419 | 0.474 |
| Style 3    | 46         | 0.0621 | 0.673 |
| All styles | 1,046      | 0.0974 | 0.564 |

The five best trials are within 0.0024 of each other, which is within the noise of a single seed, so the ranking among them is not reliable. All five use batch size 8, base width 64, and style embedding 32. The best  $\lambda_{L1}$  (199.7) sits at the top of its range, so the true optimum may lie higher. The selected configuration is trial 25, and it is used for both conditioning options.

#### D. Results

The checkpoint with the lowest validation L1 (epoch 65, validation L1 0.0854, SSIM 0.584) was selected for testing. Table XIII reports the official test split (1,046 pairs) per style. L1 is measured on images in  $[0, 1]$ .

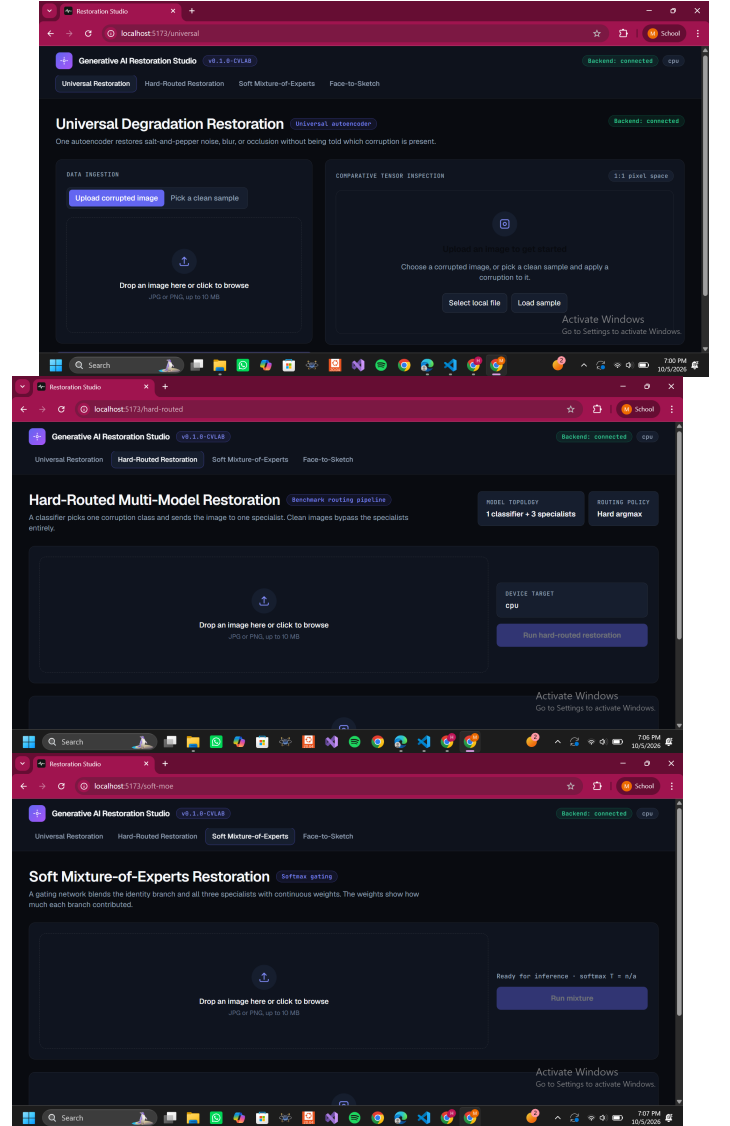
Style 2 is the hardest: its L1 is about twice that of Styles 1 and 3, and its SSIM is the lowest. Style 2 also makes up 36% of the test set, so it lowers the overall figures. Style 3 has only 46 test pairs, so its better scores are less reliable. The test L1 (0.097) is higher than the validation L1 (0.085), which is the expected generalization gap for a dataset of this size. We have not yet examined the Style 2 outputs to find the cause. One possibility is that its sketches differ more from the photographs, so the generator has to change more. The sample figure below shows the outputs at three epochs.



### VIII. APPLICATION ARCHITECTURE

The four models are served by one FastAPI backend, which validates uploads, preprocesses images, runs the ONNX models, and returns outputs with routing

information and timing. The React frontend provides four workspaces: Universal Restoration, Hard-Routed Restoration, Soft Mixture-of-Experts Restoration, and Face-to-Sketch Generator. The interface was designed in Google Stitch before it was built, and Figure 1 shows the four screens. The implemented endpoints are /health, /samples, /universal-restoration, /hard-routing, /soft-mixture, and /face-to-sketch. Each returns the restored or generated image as PNG with its inference time. An endpoint whose model is not loaded returns HTTP 503 rather than failing.



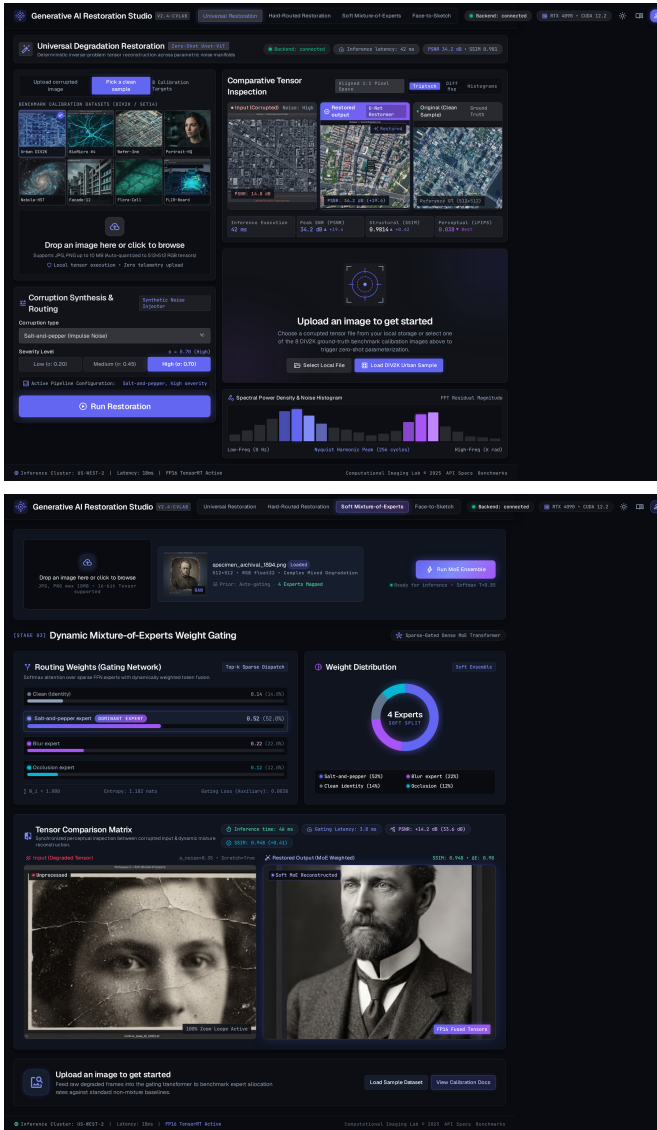
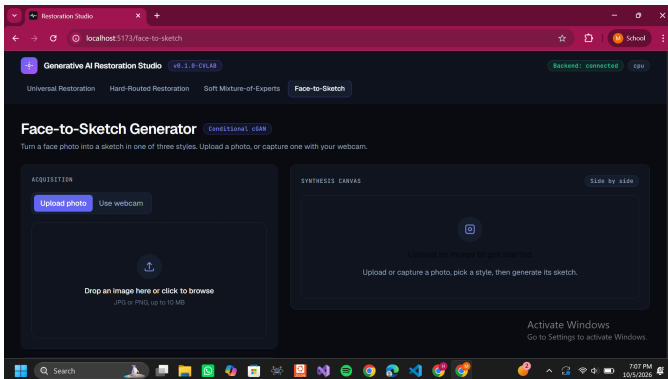
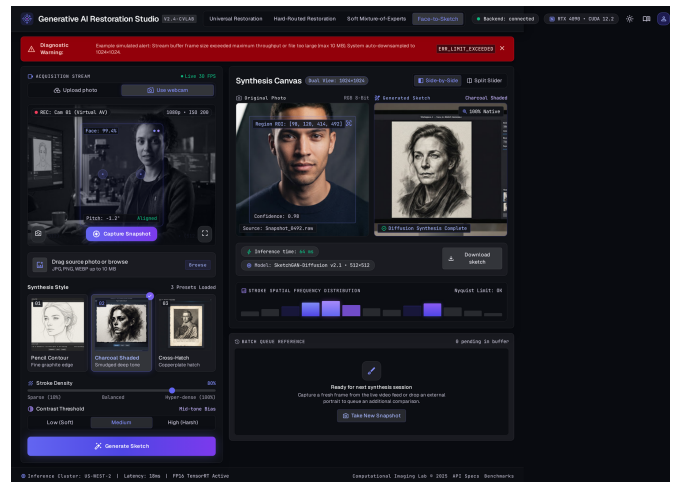
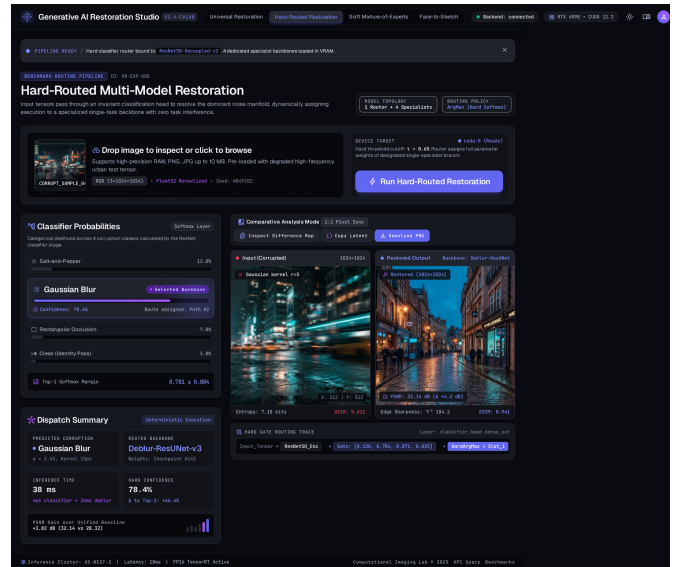


Fig. 1. Google Stitch designs for the four workspaces. The implemented interface follows these layouts.



The application is packaged with Docker Compose. It defines two services: the FastAPI backend on port 8000, and the



React frontend, served by Nginx on port 8080. The backend mounts the model folder read-only, so the trained ONNX models are supplied separately. The frontend is built with the backend's address. The whole application starts with docker compose up --build.

## IX. LIMITATIONS

- Optuna searches used short trials of two to three epochs, so the chosen hyperparameters are estimates. Full-schedule results may differ.
- Each experiment was run once with one seed. Differences of a few percent should not be over-interpreted.
- The GAN training may show decreasing losses while sample quality worsens. We rely on validation L1 and

SSIM plus visual inspection.

- FS2K is small (899 training pairs), which limits how much a GAN can learn.

## X. CONCLUSION

We built four generative systems and evaluated each one on held-out test data. The universal autoencoder shows the main limitation of this study. Its  $8 \times 8 \times 256$  bottleneck learns a smooth, generic reconstruction. It helps on occlusion, but on salt-and-pepper noise a  $3 \times 3$  median filter does better, and on blur the restored output is worse than the unrestored input. The classifier is accurate (98.7%), and the soft mixture reproduces the same behaviour: it routes salt-and-pepper to its expert but treats mild blur as clean. The face-to-sketch generator produces recognisable sketches, with a test L1 of 0.097, and Style 2 is noticeably harder than Styles 1 and 3.

The main limitations are the short Optuna searches, single seeds, and the 300-image evaluation subset used for the soft mixture. Future work would test a larger bottleneck or a perceptual loss for the autoencoder, evaluate the soft mixture on the full test set, and examine whether Style 2 needs a different conditioning approach.

## REPRODUCIBILITY

Source code: <https://github.com/Harris-05/Generative-AI-Restoration-Suite/>. Demonstration video (YouTube, unlisted): <https://youtu.be/6Q2rip2AkxE>. Model weights are available through the download links in the repository README.

## APPENDIX

### AI-USE APPENDIX

| Tool               | Used for                                                                                                     | How outputs were checked                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| ChatGPT (OpenAI)   | researching and explaining the technologies used (for example ONNX, MLflow, GroupNorm, PatchGAN, and Optuna) | explanations were checked against the original papers and official documentation                                  |
| Claude (Anthropic) | research discussion, code drafts, debugging, report drafting                                                 | every script was run on the real dataset; numbers were taken from logs; citations need checking before submission |

## REFERENCES

- [1] P. Vincent, H. Larochelle, I. Lajoie, Y. Bengio, and P.-A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. ICML*, 2008.
- [2] A. Odena, V. Dumoulin, and C. Olah, "Deconvolution and checkerboard artifacts," *Distill*, 2016.
- [3] H. Zhao, O. Gallo, I. Frosio, and J. Kautz, "Loss functions for image restoration with neural networks," *IEEE Trans. Computational Imaging*, 2017.
- [4] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE Trans. Image Processing*, 2004.
- [5] Y. Wu and K. He, "Group normalization," in *Proc. ECCV*, 2018.
- [6] D. Ulyanov, A. Vedaldi, and V. Lempitsky, "Instance normalization: The missing ingredient for fast stylization," *arXiv:1607.08022*, 2016.
- [7] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Computation*, 1991.
- [8] N. Shazeer et al., "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," in *Proc. ICLR*, 2017.
- [9] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. CVPR*, 2017.
- [10] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, "FiLM: Visual reasoning with a general conditioning layer," in *Proc. AAAI*, 2018.
- [11] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. KDD*, 2019.
- [12] O. M. Parkhi, A. Vedaldi, A. Zisserman, and C. V. Jawahar, "Cats and dogs," in *Proc. CVPR*, 2012.
- [13] D.-P. Fan et al., "FS2K: Facial sketch synthesis dataset," 2021.